Supporting Authentic Assessment in the Age of Generative AI: Evaluating Lightweight Models for AI-Generated Text Detection

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Abstract

Generative AI challenges educational assessment by enabling text that can no longer be assumed to reflect unassisted student learning. This study evaluates whether lightweight machine learning models can support educators in identifying potentially AI-generated submissions. Using 27,301 labelled essays, we compare Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine (SVM), and Random Forest classifiers with TF-IDF features. All models exceed 97% accuracy on the held-out test set, with Linear SVM achieving 99.73% accuracy and a 0.9967 F1 score in 0.76 seconds of CPU-only training. Feature analysis identifies linguistic patterns distinguishing the two classes. Rather than replacing educator judgement, we position lightweight detection as a screening tool that complements authentic, process-oriented assessment. The findings highlight the potential and limitations of accessible AI detection for resource-constrained educational environments. Source code, including data processing, training, evaluation, and feature analysis, is available on GitHub.

Keywords: AI text detection, lightweight machine learning, TF-IDF, Linear SVM, academic integrity, educational technology

Introduction

The rapid advancement of Large Language Models (LLMs) has transformed how knowledge is produced, accessed, and communicated, with generative AI increasingly becoming part of educational environments. While these technologies offer opportunities to support learning and knowledge creation, they also challenge established approaches to assessment. Students can now generate essays, code, explanations, and other academic outputs that were traditionally used as evidence of individual knowledge and competence. Consequently, educators face increasing difficulty in determining whether submitted work reflects a student's own understanding and capabilities or substantial assistance from intelligent systems. This shift has intensified concerns surrounding academic integrity and has prompted calls for assessment approaches that remain meaningful and authentic when generative AI is widely available. Rather than treating AI detection as a replacement for assessment, there is a growing need to consider how intelligent technologies can support educators in evaluating student work while retaining human judgement and learning-centred assessment practices.

One response to this challenge is the development of AI-generated text (AIGT) detection methods that can provide educators with additional evidence when reviewing student submissions. Existing research has explored a range of approaches, including watermarking, statistical and stylistic analysis, machine learning classifiers, and transformer-based models. Fraser et al. (2025) provided a comprehensive survey of state-of-the-art AIGT detection approaches and identified several factors influencing detectability across different generation and evaluation scenarios. Prova (2024) demonstrated that machine learning approaches, including XGB Classifier, Support Vector Machine (SVM), and BERT, can distinguish between AI-generated and human-written text, reporting accuracies of 0.84, 0.81, and 0.93, respectively. Widjaya and Iskandar (2026) further demonstrated the effectiveness of traditional machine learning by combining TF-IDF representations with an SVM classifier on a large dataset of 487,235 text samples, achieving high accuracy and macro F1 scores. Similarly, Pantic and Mugosa (2025) investigated Random Forest with TF-IDF features for AI-generated content detection in the public health domain, reporting 81.8% accuracy and highlighting the influence of domain characteristics on detection performance.

Although transformer-based approaches can provide strong detection performance, their deployment may introduce practical challenges for educational institutions, including computational requirements, dependence on specialised infrastructure, limited interpretability, and vulnerability to changes in text-generation and paraphrasing strategies. Rehman et al. (2026), for example, proposed a Retrieval-Augmented Generation (RAG)-inspired framework combined with a fine-tuned transformer classifier and achieved 97.53% accuracy while incorporating LIME-based explainability. Such approaches demonstrate the potential of sophisticated models but also motivate investigation into whether less computationally demanding alternatives can provide sufficiently reliable signals for educational assessment contexts. This question is particularly relevant for educators and institutions operating with limited computational resources, where deploying and maintaining large transformer-based systems may not always be practical.

Despite the growing body of research on AIGT detection, comparatively less attention has been given to the role of lightweight detection models as accessible supporting technologies within educational assessment workflows. In particular, there is a dire need to understand whether conventional machine learning approaches can provide strong detection performance while requiring minimal computational infrastructure. Addressing this practical gap is important because an accessible detection mechanism could allow educators to incorporate AI-generated-text analysis into broader assessment practices without requiring dedicated GPU infrastructure or complex deployment environments. However, such systems should be viewed as decision-support tools rather than autonomous mechanisms for determining authorship or academic misconduct.

Building on this gap, this study empirically compares four lightweight machine learning classifiers—Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine, and Random Forest—using TF-IDF representations to distinguish AI-generated from human-written essays. The models are evaluated on a dataset of 27,301 labelled essays using standard classification metrics and training time on a standard laptop without GPU acceleration. The study therefore addresses the following research question:

RQ1: To what extent can lightweight machine learning models accurately distinguish AI-generated from human-written student essays while remaining computationally accessible for educational assessment contexts?

The study contributes to the Information Systems education literature by providing empirical evidence on the feasibility of lightweight intelligent technologies as supporting components of AI-aware assessment. Beyond classification performance, the study examines the linguistic features associated with the two classes and discusses how such technologies can complement, rather than replace, educator judgement and authentic assessment practices. In doing so, the study responds directly to the theme of **"Assessment in the Age of Intelligent Systems"**, particularly the need to rethink assessment practices and explore how intelligent systems can support educators when AI-generated outputs are increasingly available. The findings also highlight the importance of using AI detection cautiously within broader assessment strategies that may include process-based evaluation, oral explanation, reflection, experiential activities, and other forms of evidence of student learning.

Methodology

Dataset and Preprocessing

We used the *TrainingEssayData* corpus, comprising 27,301 labelled essays written by humans or generated by AI systems. The dataset contains 16,117 human-written essays (59.0%) and 11,184 AI-generated essays (41.0%), representing a moderate class imbalance. The essays cover diverse writing contexts, including personal narratives, argumentative writing, and analytical responses. To prepare the data, we removed missing records, converted text to a consistent string format, trimmed leading and trailing whitespace, excluded essays shorter than 20 characters, and removed exact duplicate texts. Labels were encoded as binary values, with 0 representing human-written text and 1 representing AI-generated text.

Feature Representation and Classification Models

We represented each essay using Term Frequency–Inverse Document Frequency (TF-IDF), which weights terms according to their importance within individual documents and across the corpus. The vectorizer was configured with a maximum of 20,000 features, unigram and bigram representations, a minimum document frequency of 3, and sublinear term-frequency scaling. The resulting representation was a sparse 20,000-dimensional feature matrix.

Four lightweight machine learning classifiers were evaluated. **Logistic Regression (LR)** provided a regularized linear baseline suitable for high-dimensional sparse text representations. **Multinomial Naive Bayes (NB)** provided a computationally efficient probabilistic baseline. **Linear Support Vector Machine (SVM)** was selected because of its suitability for sparse, high-dimensional text features and its strong performance in text classification. **Random Forest (RF)** was included as a non-linear ensemble model capable of capturing feature interactions. Class weighting was applied to LR, SVM, and RF to account for the moderate class imbalance.

Experimental Design and Evaluation

The dataset was divided into training (70%), validation (15%), and test (15%) subsets using stratified random sampling, resulting in 19,110 training, 4,095 validation, and 4,096 test essays. The validation set was used during model development, while the test set was reserved for final performance evaluation. Models were assessed using accuracy, precision, recall, F1 score, and ROC-AUC. Training time was additionally recorded to evaluate computational requirements.

All experiments were conducted on a standard laptop with 16 GB RAM and without GPU acceleration. This setup was selected to reflect the computational conditions potentially available to educators and institutions without dedicated AI infrastructure. The experimental design therefore evaluates not only classification performance but also the practical accessibility of lightweight detection approaches.

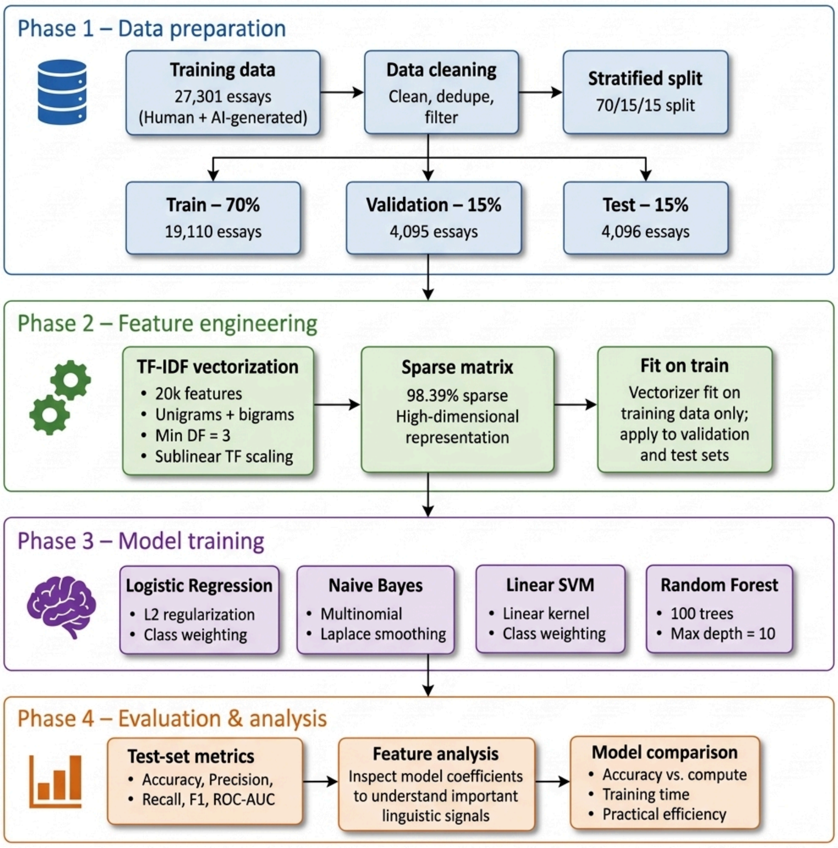

Figure 1. Methodology Flowchart for AI-Generated Text Detection.

The flowchart illustrates the overall experimental pipeline, from dataset preprocessing and TF-IDF feature extraction to model training, evaluation, and linguistic feature analysis.

Result

Classification Performance

The four lightweight classifiers achieved strong performance on the held-out test set, with accuracy exceeding 97% across all models (Table 1). Linear SVM achieved the highest overall performance, obtaining 99.73% accuracy, 99.82% precision, 99.52% recall, and a 99.67% F1 score, with an ROC-AUC of 1.000. This corresponds to 11 misclassified essays out of 4,096 test samples.

Logistic Regression also performed strongly, achieving 98.93% accuracy and a 98.68% F1 score. Multinomial Naive Bayes achieved 97.53% accuracy and a 96.95% F1 score, while Random Forest achieved 97.41% accuracy and a 96.81% F1 score. All models produced ROC-AUC values above 0.99, indicating strong class discrimination on the evaluated test set.

These results in Figure 2 indicate that conventional lightweight classifiers combined with TF-IDF representations can achieve high classification performance on the evaluated educational essay corpus. The exceptionally high performance of Linear SVM should, however, be interpreted in the context of this dataset and experimental setting rather than as evidence of universal AI-generated text detectability. The comparable validation (99.56%) and test (99.73%) accuracies, together with the removal of exact duplicate texts, provide no immediate indication of substantial overfitting to the test partition. Nevertheless, the observed performance may partly reflect dataset-specific lexical, stylistic, topical, or collection-related patterns. Therefore, these results demonstrate the feasibility of lightweight AI-text classification on the evaluated corpus, while generalisability to unseen domains, prompts, generation systems, and paraphrased or adversarially modified text requires further investigation.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Training Time (s)
LG	0.9893	0.9910	0.9827	0.9868	0.9995	0.39
NB	0.9753	0.9823	0.9571	0.9695	0.9943	0.02
Linear SVM	0.9973	0.9982	0.9952	0.9967	1.000	0.76
RF	0.9741	0.9781	0.9583	0.9681	0.9940	1.42

Table 1. Model Performance Comparison.

The table compares the four lightweight classifiers across classification performance and computational efficiency, showing that Linear SVM provides the strongest overall performance while requiring less than one second for training.

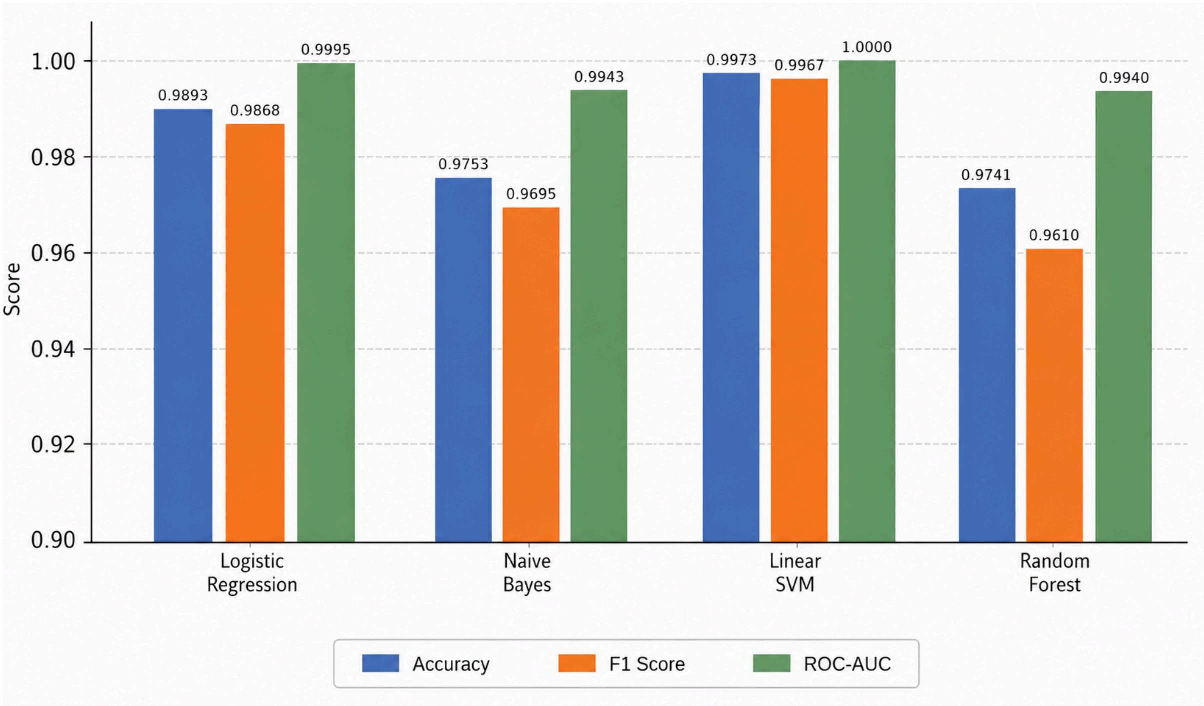

Figure 2. Model Comparison.

The figure provides a visual comparison of the classification performance of the four models, highlighting the consistently strong results and the superior performance of Linear SVM across the evaluated metrics.

Linguistic Feature Analysis

To examine the linguistic signals used by the best-performing classifier, we analysed the Linear SVM coefficients. Table 2 presents features with the largest positive and negative weights, representing terms that were most strongly associated with the AI-generated and human-written classes, respectively.

Several terms associated with the AI-generated class included transitional or hedging expressions such as “additionally,” “often,” and “potential.” In contrast, features associated with the human-written class included more topic-specific terms such as “Venus,” “driving,” and “car,” alongside terms such as “because,” “people,” and “very.” These patterns suggest that the classifier may be exploiting differences in lexical and stylistic distributions within the dataset.

Importantly, these features should not be interpreted as universal indicators of AI authorship. A particular word or linguistic pattern cannot independently establish whether a student has used generative AI. Rather, the analysis provides interpretable evidence of the linguistic signals contributing to classification within the evaluated corpus.

AI-Generated Indicator	Weight	Human-written Indicator	Weight
it not	1.546	because	-2.912
it like	1.433	venus	-1.654
and	1.216	driving	-1.533
potential	1.207	would	-1.419
to	1.192	car	-1.306
often	1.156	the students	-1.263
additionally	1.149	very	-1.198
important to	1.054	people	-1.184
example if	1.005	dont	-1.140
important	0.961	because of	-1.063

Table 2. Top Discriminative Features for Linear SVM.

The table presents the highest-weighted TF-IDF features associated with AI-generated and human-written classes, providing insight into the linguistic patterns used by the Linear SVM classifier.

AI-generated text is characterized by frequent use of transitional and hedged language. The features "additionally," "often," and "potential" reflect the stylistic patterns of AI text generation, which tend to employ predictable connectors and cautious language. The inclusion of common words like "and" and "to" as positive indicators suggests subtle differences in syntactic patterns, with AI text potentially using these connectors more frequently to generate fluent text. Human-written text is distinguished by more diverse and specific vocabulary. Features such as "venus," "driving," and "car" indicate the inclusion of concrete examples and specific topics, while "because," "very," "people," and informal contractions like "dont" reflect more casual, personal, and varied writing styles (Figure 3).

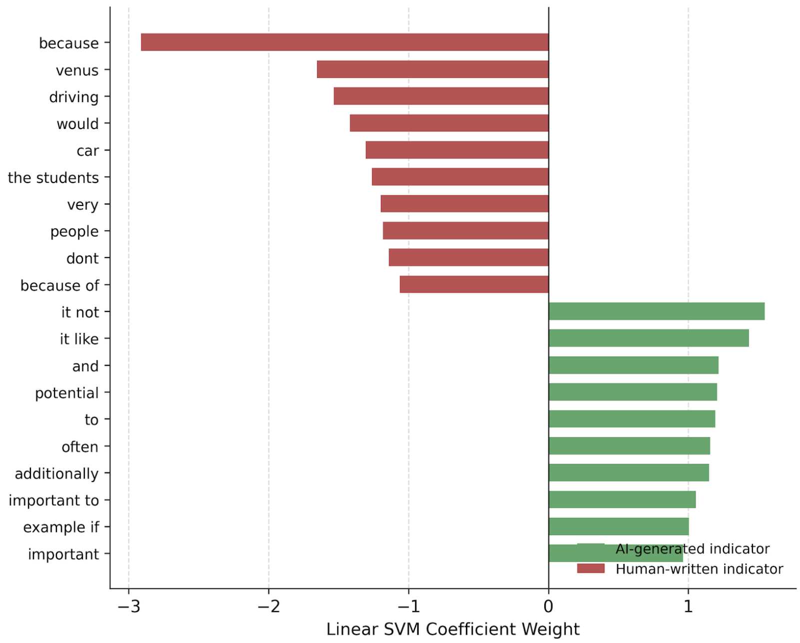

Figure 3. Feature Weights.

The figure visualises the relative importance of the most discriminative features, illustrating how specific lexical and stylistic patterns contribute to the Linear SVM's classification decisions.

Computational Efficiency

The models required between 0.02 and 1.42 seconds for training on the standard laptop environment. Multinomial Naive Bayes was the fastest model at 0.02 seconds, followed by Logistic Regression at 0.39 seconds and Linear SVM at 0.76 seconds. Random Forest required the longest training time at 1.42 seconds.

Although these measurements are hardware- and implementation-dependent, the results demonstrate that all four approaches can be trained with minimal computational resources and without GPU acceleration. This characteristic is particularly relevant to educational environments where access to specialised AI infrastructure may be limited. The findings therefore suggest that lightweight models can provide a technically accessible option for incorporating AI-generated-text analysis into assessment-support workflows.

Discussion and Educational Implications

The results suggest that lightweight machine learning models can provide strong AI-generated-text classification performance on the evaluated educational essay corpus without requiring GPU infrastructure. However, the role of such tools in education should be supportive rather than determinative. A lightweight classifier could serve as an initial screening mechanism, prompting educators to consider additional evidence of learning such as drafts, revision processes, oral explanations, or reflections rather than treating a detection score as evidence of academic misconduct. This aligns with the conference theme of *Assessment in the Age of Intelligent Systems*, where assessment practices increasingly need to combine intelligent technologies with authentic, process-based evaluation. The low computational requirements also suggest that such approaches could be accessible to institutions with limited technical infrastructure. Nevertheless, the dataset-specific nature of the results, potential false positives, evolving LLMs, and paraphrasing remain important considerations, highlighting the need for human judgement and responsible use of AI detection in educational assessment.

For educators, administrators, and university assessment teams, these findings carry a practical implication beyond classification accuracy: a lightweight TF-IDF-based classifier can be trained and deployed in under one second on ordinary laptop hardware, without GPU infrastructure, cloud subscriptions, or paid third-party detection services. This lowers the barrier for institutions with limited technology budgets to incorporate an initial AI-text screening step into existing assessment workflows, rather than relying on commercial detectors whose licensing costs, opacity, and data-handling practices may be difficult for smaller institutions to justify or audit. Because the approach requires only standard machine learning libraries and modest compute, universities can run and update such screening tools in-house, retaining control over how flagged submissions are reviewed and ensuring that detection remains one input among several rather than the sole basis for an academic integrity decision.

For educational researchers, the study's transparent methodology, documented preprocessing pipeline, and openly released source code provide a reusable starting point rather than an isolated result. Because the pipeline is not tied to a particular language, generation system, or writing genre, it offers a template that future work could adapt to multilingual AI-text detection, where labelled corpora and detection benchmarks remain comparatively scarce. The same codebase could also serve as a lightweight baseline against which more computationally demanding deep learning and large language model-based detectors are benchmarked, helping researchers quantify the marginal gains such approaches offer over simple, interpretable classifiers before committing to their additional computational and infrastructural cost.

Limitations and Future Research

This study has several limitations. First, the evaluation is based on a single labelled essay corpus, and the high classification performance may partly reflect dataset-specific linguistic, topical, or source characteristics. Therefore, the results should not be generalised to all student writing or current and future generative AI systems. Second, the study evaluates conventional AI-generated versus human-written text and does not explicitly examine mixed-authorship writing, human-edited AI output, or extensive paraphrasing. Third, computational measurements were obtained from a single laptop environment and should therefore be interpreted as indicative rather than universal benchmarks.

Future research should evaluate the models across multiple educational datasets, disciplines, writing contexts, and generations of LLMs. Further investigation of paraphrasing, human-AI collaborative writing, and adversarial transformations is also required. Importantly, future studies should examine educators' and students' perceptions of AI detection and investigate how detection technologies can be integrated into assessment practices without encouraging excessive surveillance or undermining trust. Such work would strengthen understanding of how lightweight intelligent systems can responsibly support assessment in AI-augmented educational environments.

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